GRADE 7 ACCELERATED · UNIT 5 · LESSON 1

Scale Drawings

Question Bank - Additional 5.01 **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

For questions 1 through 4, find the scale factor from the original to the copy.

1. A 3 cm side becomes 12 cm.

k = 4

2. A 10 cm side becomes 2 cm.

k = $\frac{1}{5}$

3. A 3 cm by 4 cm rectangle is copied at 4.5 cm by 6 cm.

k = 1.5

4.5 ÷ 3 and 6 ÷ 4 both give 1.5.

4. A 20 cm side becomes 15 cm.

k = ¾

For questions 5 through 8, use the scale factor or the scale.

5. The scale factor is 3 and a side of the original is 3.5 cm. How long is that side in the copy?

10.5 cm

6. The scale factor is 0.25 and a side of the original is 12 cm. How long is that side in the copy?

3 cm

7. A map's scale is $1\text{ in} : 40\text{ mi}$. Two cities are 3.25 in apart on the map. How far apart are they?

130 mi

8. On the same map, a river is 90 mi long. How long is it on the map?

2.25 in

Back to the map: divide by 40.

Choose one.

9. A drawing has a 30° angle. The real figure is a scale copy at $k = 4$. What is the angle in the real figure?

- ○ A. 34°
- ○ B. 120°
- ○ C. 7.5°
- ● **D. 30°**

D

A scale factor multiplies lengths; angles do not change.